Experiment: Automate application deployment using Jenkins

Weight: 5 marks | Course outcome: CO1

Deployment was automated end to end: Jenkins polls GitHub for new commits, and after a successful build it deploys the app to /opt/devops-lab1, restarts the systemd service and verifies the health endpoint. A pushed commit (bump app version to 1.0.1) triggered build #2 automatically without any manual action.

## Build trigger

- Poll SCM with schedule H/5 * * * * - Jenkins checks GitHub every ~5 minutes
- When a new commit is found, the pipeline is queued automatically
- Post-build behaviour is handled by the Jenkinsfile post section (success/failure messages), artifact archiving and the deploy stage

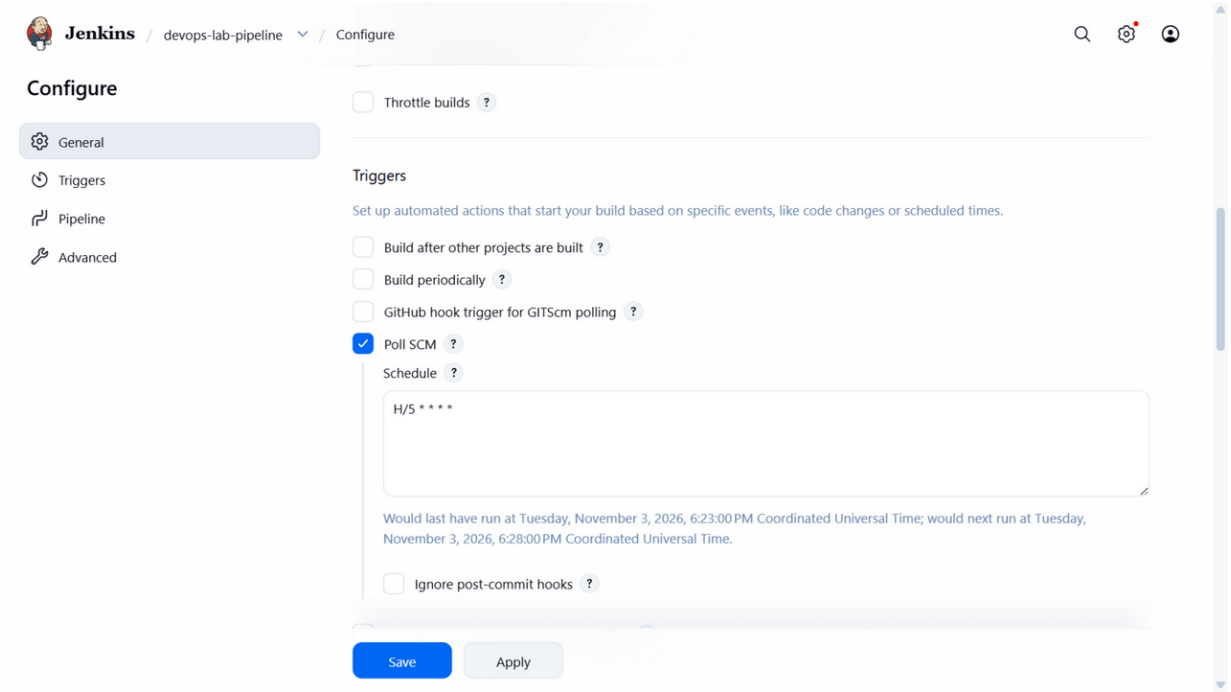



Fig: Poll SCM trigger configuration

## Deployment steps

The Deploy stage runs scripts/deploy.sh with passwordless sudo restricted to this one script through /etc/sudoers.d/jenkins-deploy:

```
jenkins ALL=(ALL) NOPASSWD: /usr/bin/bash /var/lib/jenkins/workspace/devops-lab-pipeline/scripts/deploy.sh
```

### scripts/deploy.sh

```
#!/bin/bash
# Deploys the flask app to /opt/devops-lab1 and restarts the systemd service
set -e

APP_SRC="$(cd "$(dirname "$0")/.." && pwd)"
APP_DEST="/opt/devops-lab1"

echo "Deploying from $APP_SRC to $APP_DEST ..."

sudo mkdir -p "$APP_DEST"
sudo rm -rf "$APP_DEST/app" "$APP_DEST/tests"
sudo cp -r "$APP_SRC/app" "$APP_SRC/tests" "$APP_DEST/"

cd "$APP_DEST"
if [ ! -d venv ]; then
    python3 -m venv venv
fi
./venv/bin/pip install -q -r app/requirements.txt

sudo systemctl restart devops-lab1.service
sleep 3

STATUS=$(curl -s -o /dev/null -w "%{http_code}" http://localhost:5000/health || true)
if [ "$STATUS" = "200" ]; then
    echo "Deploy successful - health check returned HTTP $STATUS"
else
    echo "Health check failed (HTTP $STATUS)"
    exit 1
fi
```

### /etc/systemd/system/devops-lab1.service

```
[Unit]
Description=DevOps Lab 1 Flask Application
After=network.target

[Service]
User=www-data
WorkingDirectory=/opt/devops-lab1
ExecStart=/opt/devops-lab1/venv/bin/python -m flask --app app.app run --host 0.0.0.0 --port 5000
Restart=on-failure

[Install]
WantedBy=multi-user.target
```

## Automatic deployment evidence (build #2)

Commit 7530c85 (bump app version to 1.0.1) was pushed to GitHub. The next poll found the change and Jenkins started build #2 on its own - the console confirms Started by an SCM change. After the deploy stage the running app reported the new version.

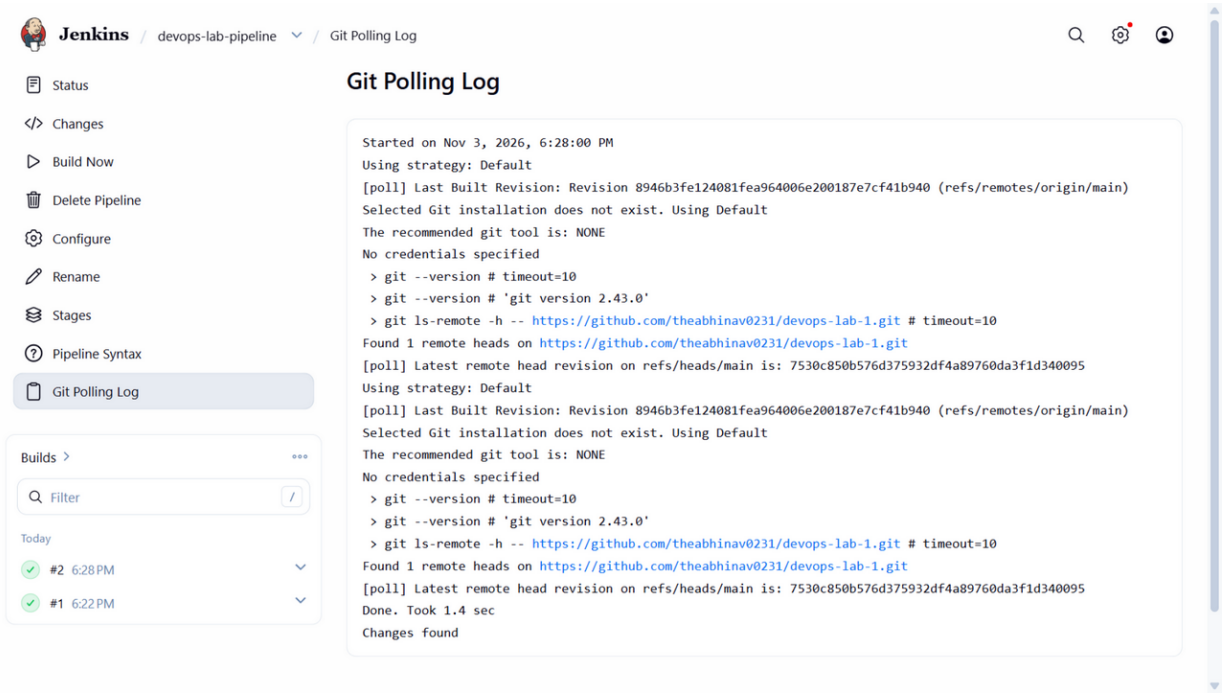



Fig: Git polling log: poll detected the new commit (Changes found)

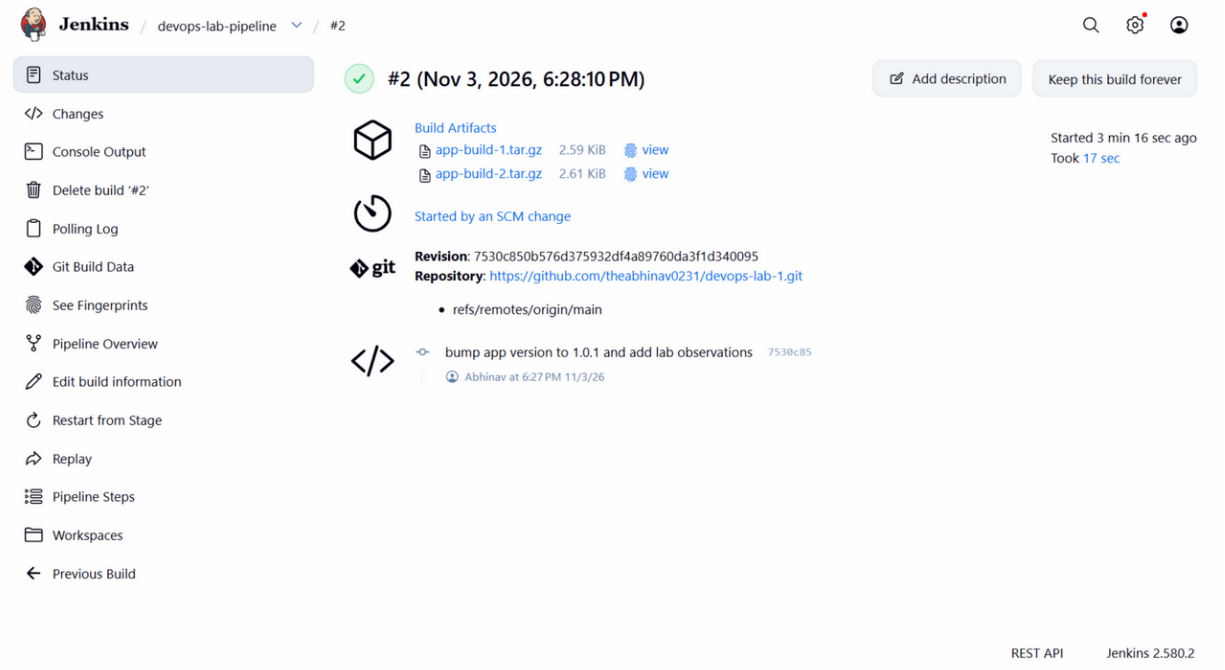



Fig: Build #2 - Started by an SCM change, revision 7530c85

```
$ curl http://localhost:5000/health   (after automatic build #2)
{"status":"UP","version":"1.0.1"}

$ git -C /var/lib/jenkins/workspace/devops-lab-pipeline log -1 --oneline
7530c85 bump app version to 1.0.1 and add lab observations
```

```
abhinav@LAPTOP-BSV7PCJK: ~/devops-lab1-ci  —  terminal
abhinav@LAPTOP-BSV7PCJK:~$ curl http://localhost:5000/health   (after automatic build #2)
{"status":"UP","version":"1.0.1"}
abhinav@LAPTOP-BSV7PCJK:~$ git -C /var/lib/jenkins/workspace/devops-lab-pipeline log -1 --oneline
7530c85 bump app version to 1.0.1 and add lab observations
abhinav@LAPTOP-BSV7PCJK:~$ head -3 of build 2 console (trigger source)
Started by an SCM change
Obtained Jenkinsfile from git https://github.com/theabhinav0231/devops-lab-1.git
[Pipeline] Start of Pipeline
```

Fig: Deployed app health check after the automatic build

```
abhinav@LAPTOP-BSV7PCJK: ~/devops-lab1-ci  —  terminal
abhinav@LAPTOP-BSV7PCJK:~$ curl http://localhost:5000/health   (after pipeline deploy)
{"status":"UP","version":"1.0.0"}
abhinav@LAPTOP-BSV7PCJK:~$ systemctl status devops-lab1 --no-pager | head -5
● devops-lab1.service - DevOps Lab 1 Flask Application
     Loaded: loaded (/etc/systemd/system/devops-lab1.service; disabled; preset: enabled)
     Active: active (running) since Tue 2026-11-03 18:22:50 UTC; 27s ago
   Main PID: 2666 (python)
      Tasks: 1 (limit: 9172)
abhinav@LAPTOP-BSV7PCJK:~$ ls /opt/devops-lab1
app
tests
venv
```

Fig: Service restarted by the pipeline deploy stage

## Automation results

| Build | Trigger | Result | Time | Deployed version |
|---|---|---|---|---|
| #1 | Build Now (manual) | SUCCESS | 29 s | 1.0.0 |
| #2 | Poll SCM (automatic) | SUCCESS | 17 s | 1.0.1 |

## Observations

- The deploy script is idempotent - it can run on every build without manual cleanup.
- The health check inside deploy.sh makes a bad deployment fail the pipeline immediately.
- sudo access for the jenkins user is limited to the single deploy script (least privilege).
- No manual step was needed between git push and the new version being served.

## Conclusion

All the commands were executed successfully on Ubuntu 24.04 LTS and the evidence (command logs, screenshots and commit history) is committed in the GitHub repository github.com/theabhinav0231/devops-lab-1 under the docs/ directory.